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(54) **ADAPTIVE RESISTANCE SYSTEM FOR PHYSICAL THERAPY AND BIOMECHANICS TRAINING**

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(52) **U.S. Cl.**

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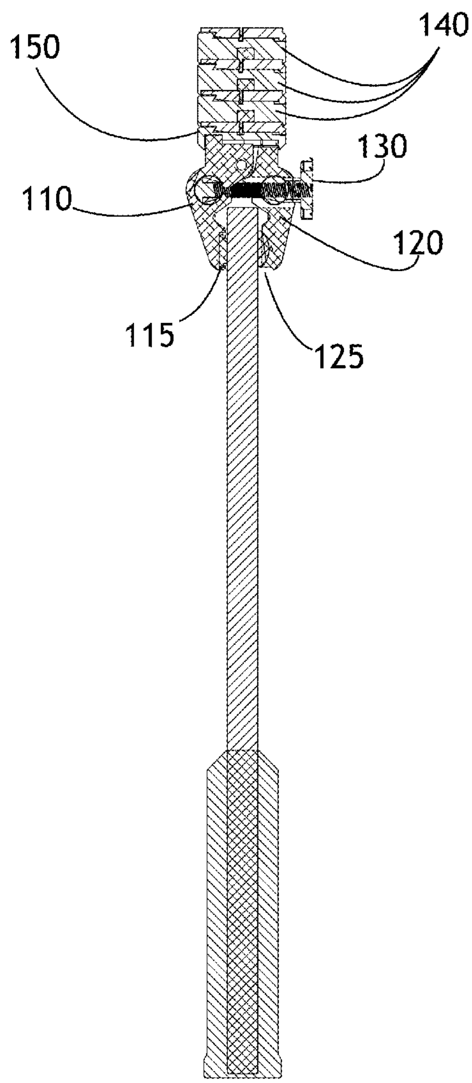
(57) **ABSTRACT**

An adaptable resistance system with improved attachment structures may feature a clamp body with modular weight discs which may then be added to the clamp body. The clamp structure allows for the use of the training weight on any position on an element of sports equipment, such as a paddle, racket, or stick, as well as the use of multiple weights on the same piece of equipment. Locking mechanisms may be employed to further secure the clamp of the resistance system to the sports equipment. Modular weights allow for further customization as a user may remove all modular weights or selectively position a desired quantity on the resistance system.



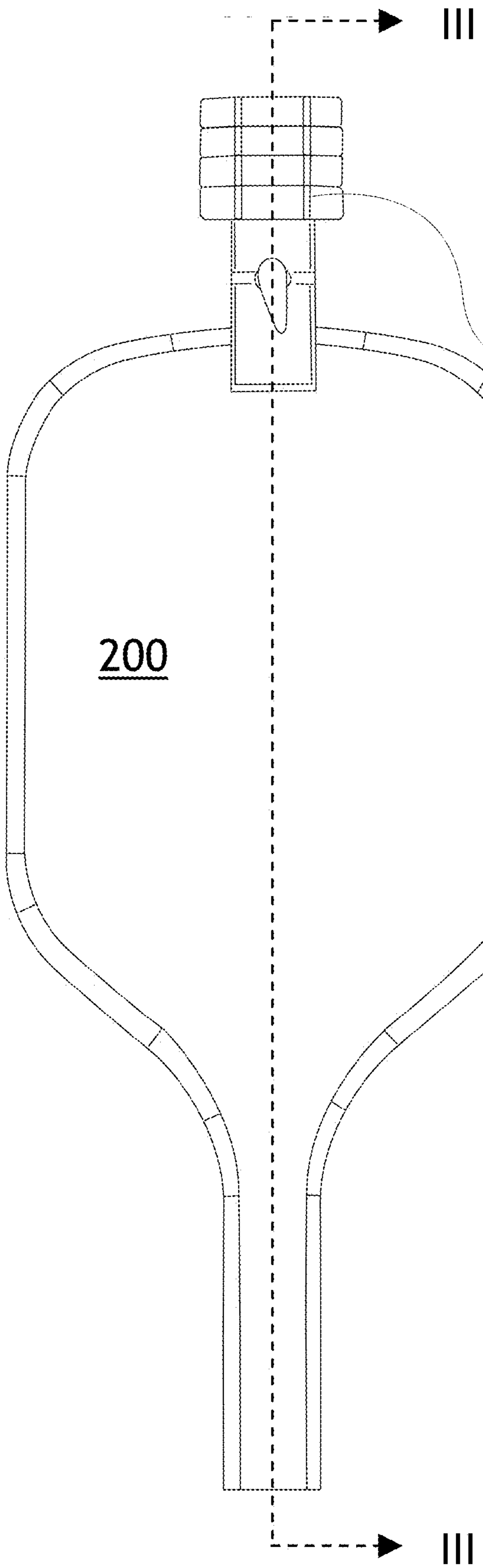


FIG. 1

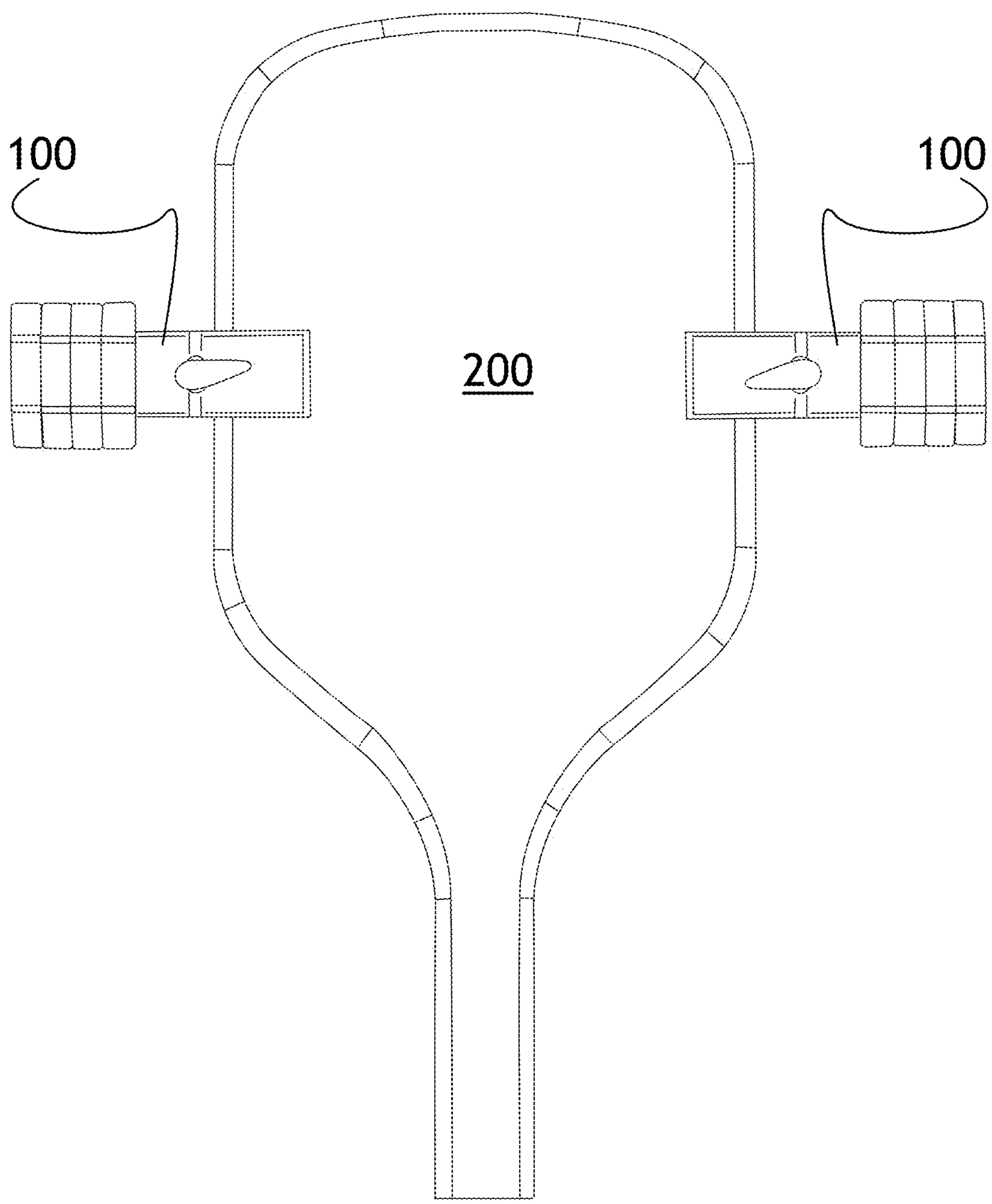


FIG. 2

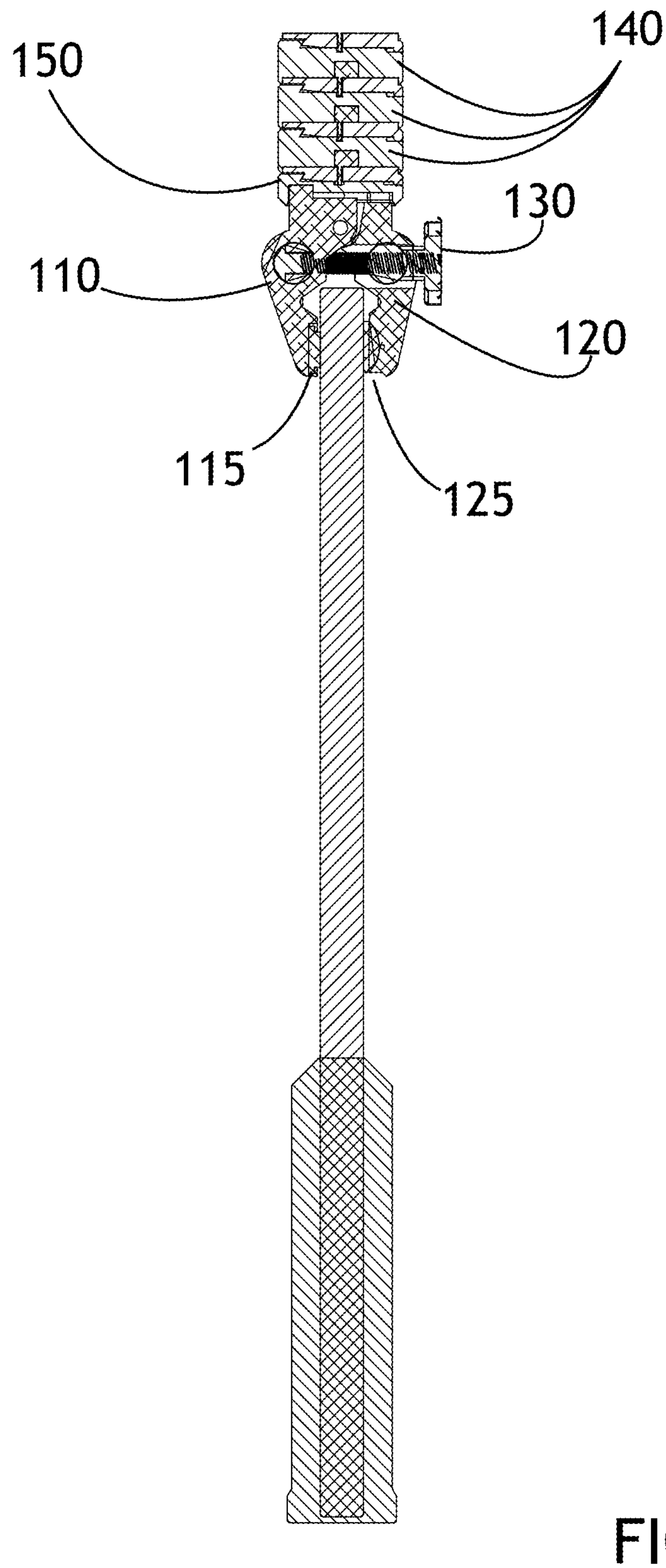


FIG. 3

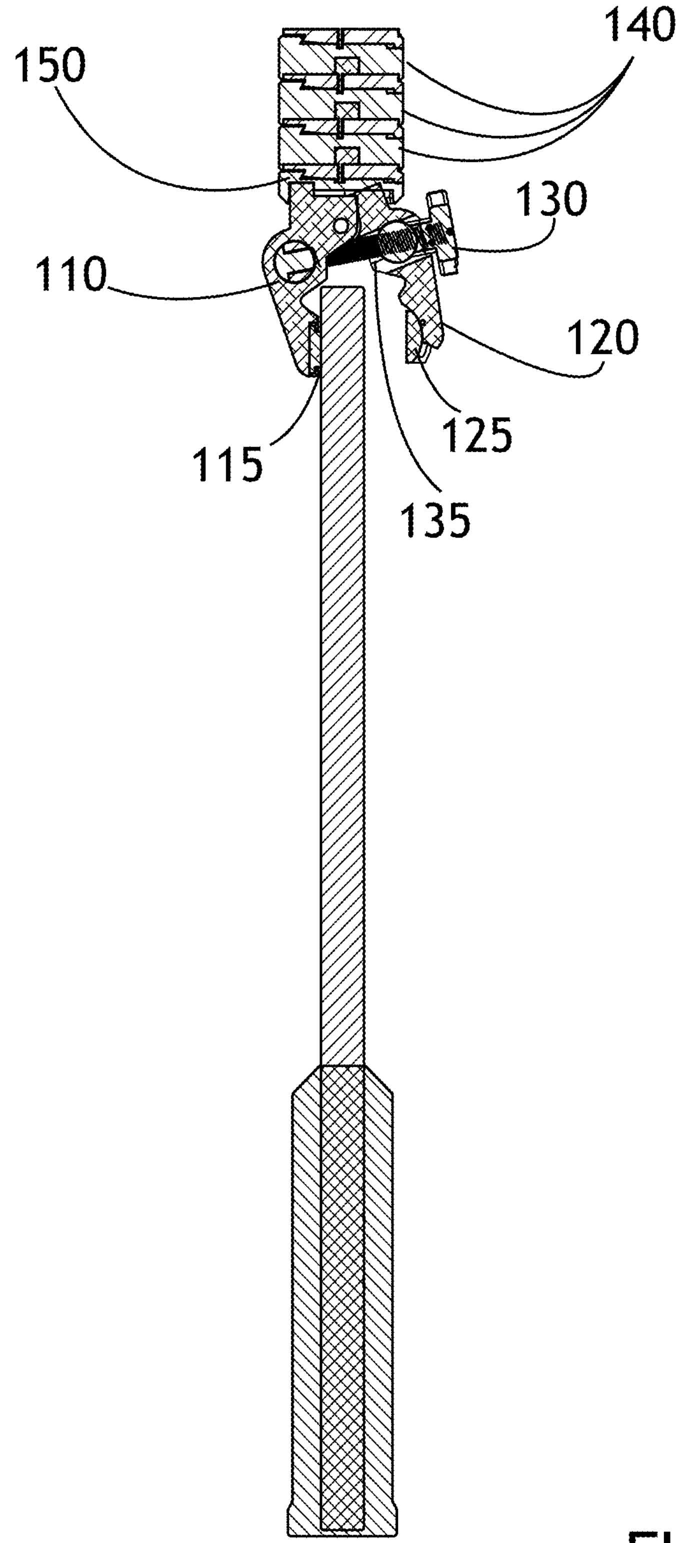


FIG. 4

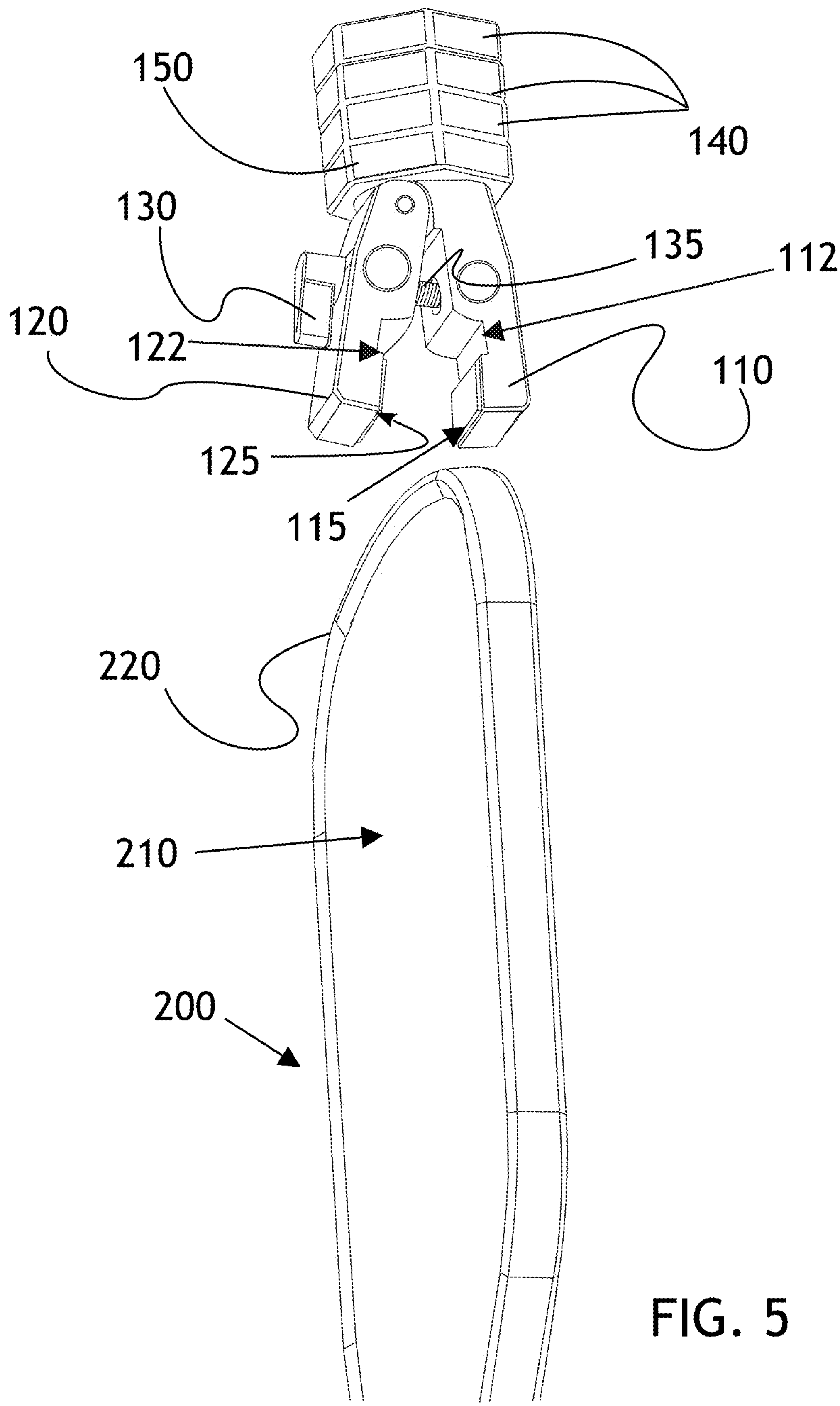


FIG. 5

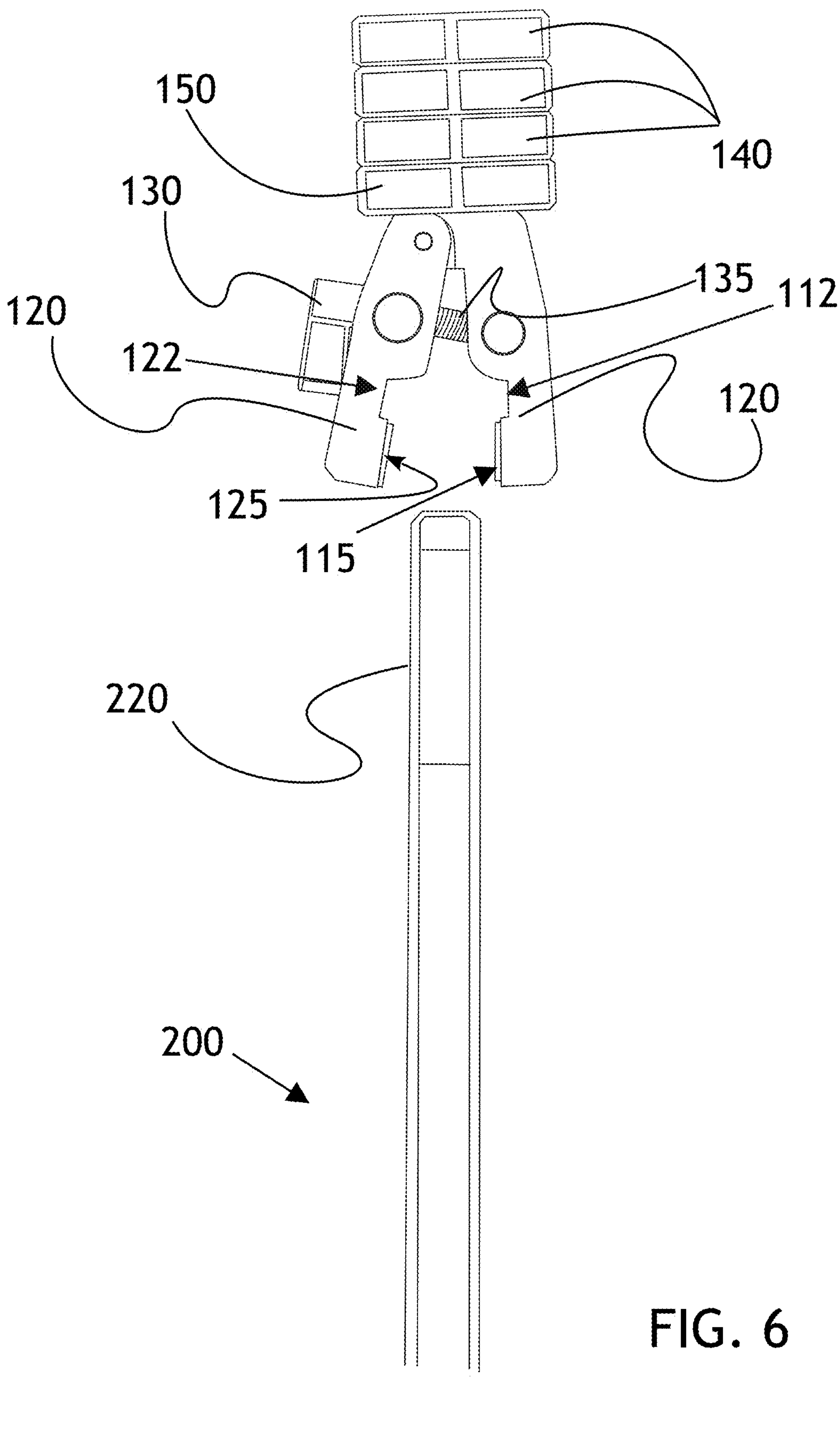


FIG. 6

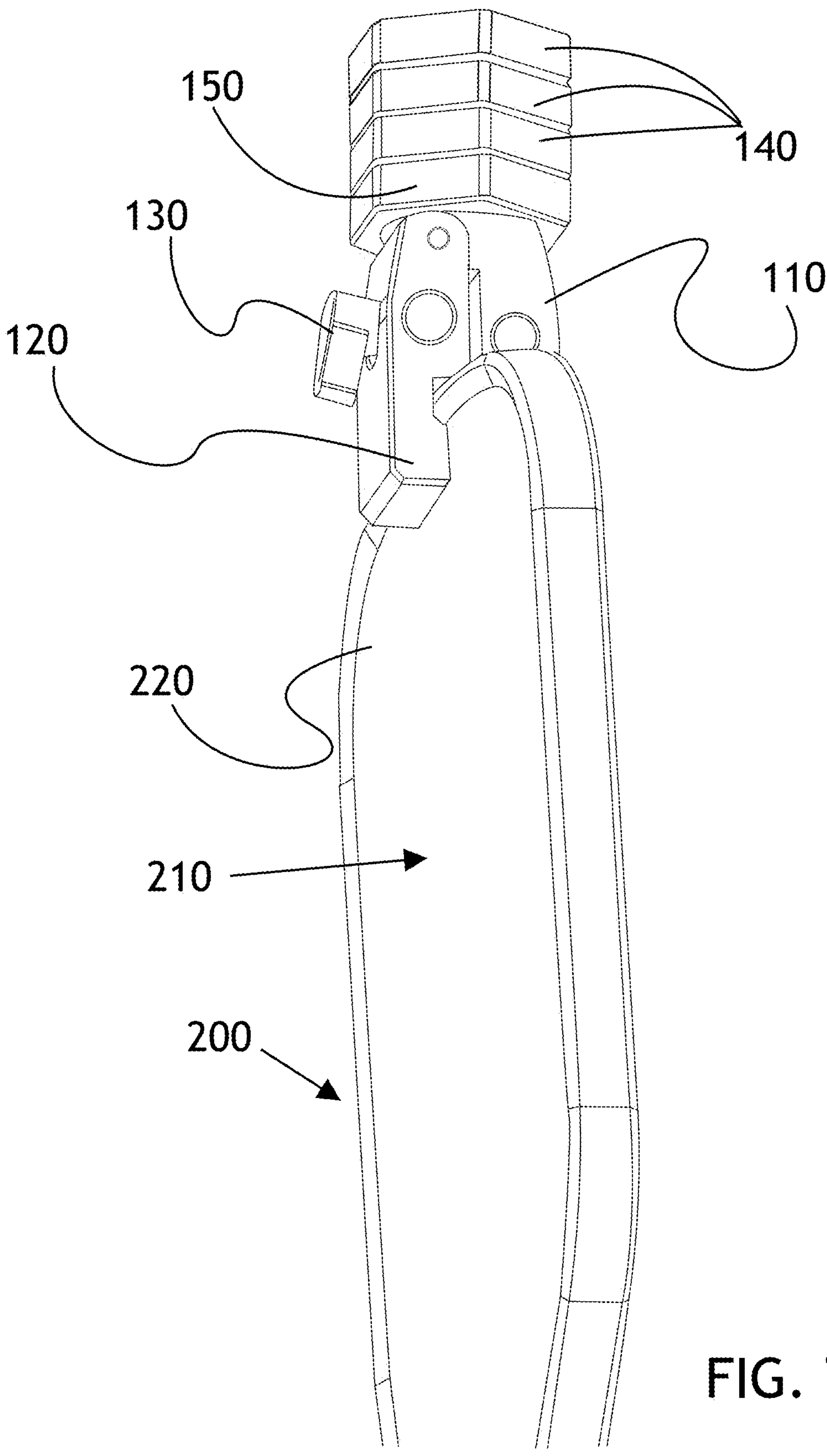


FIG. 7

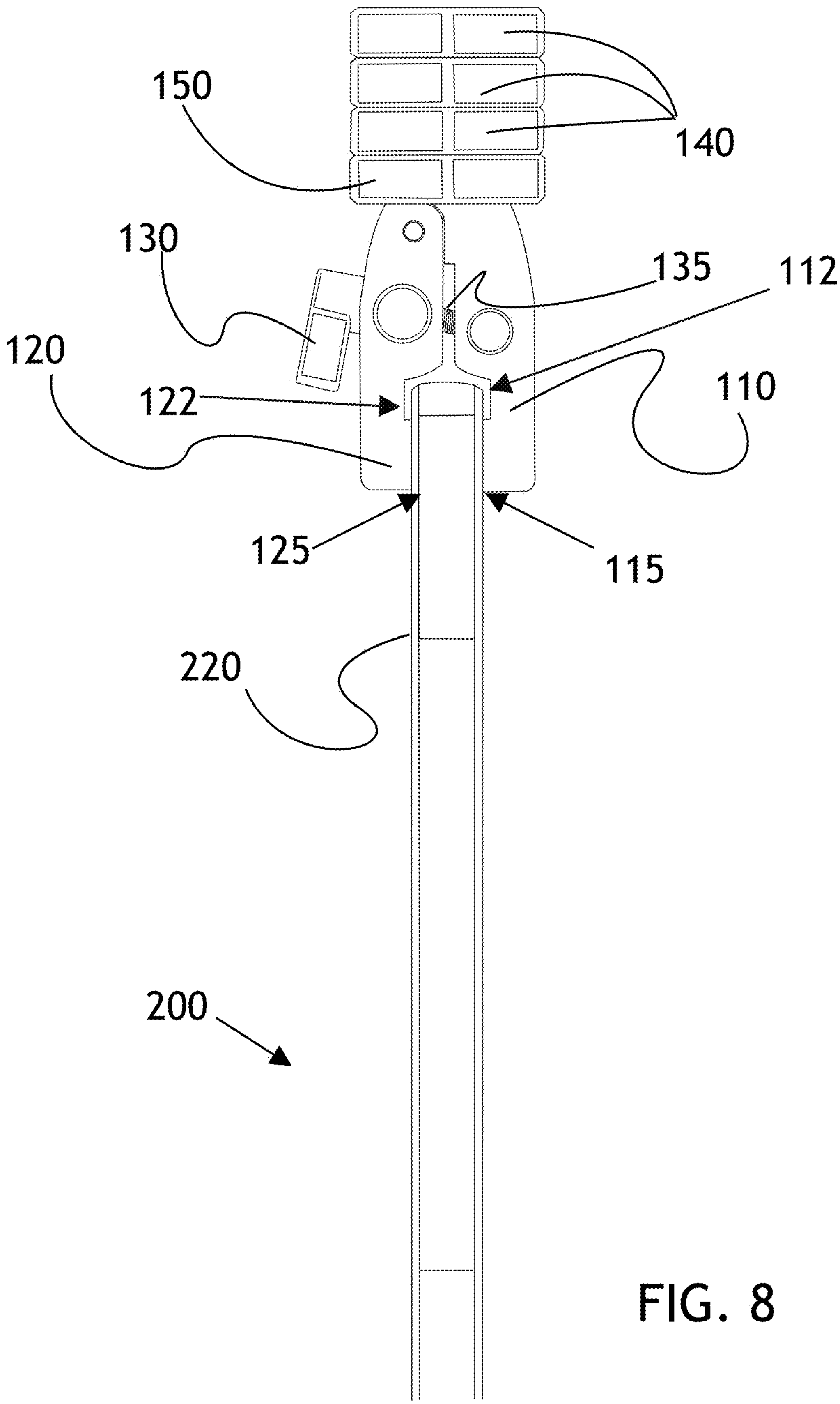


FIG. 8

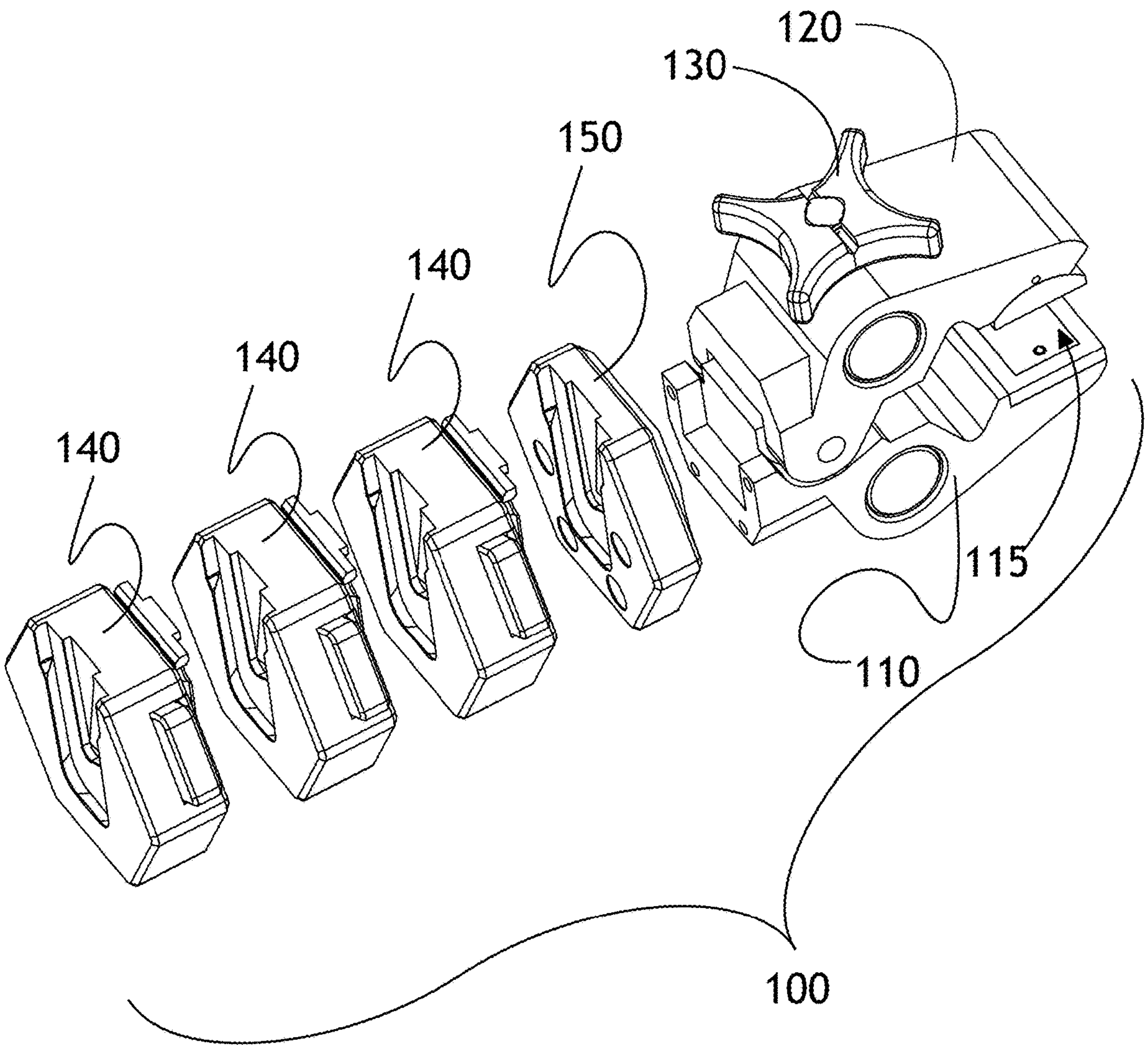


FIG. 9

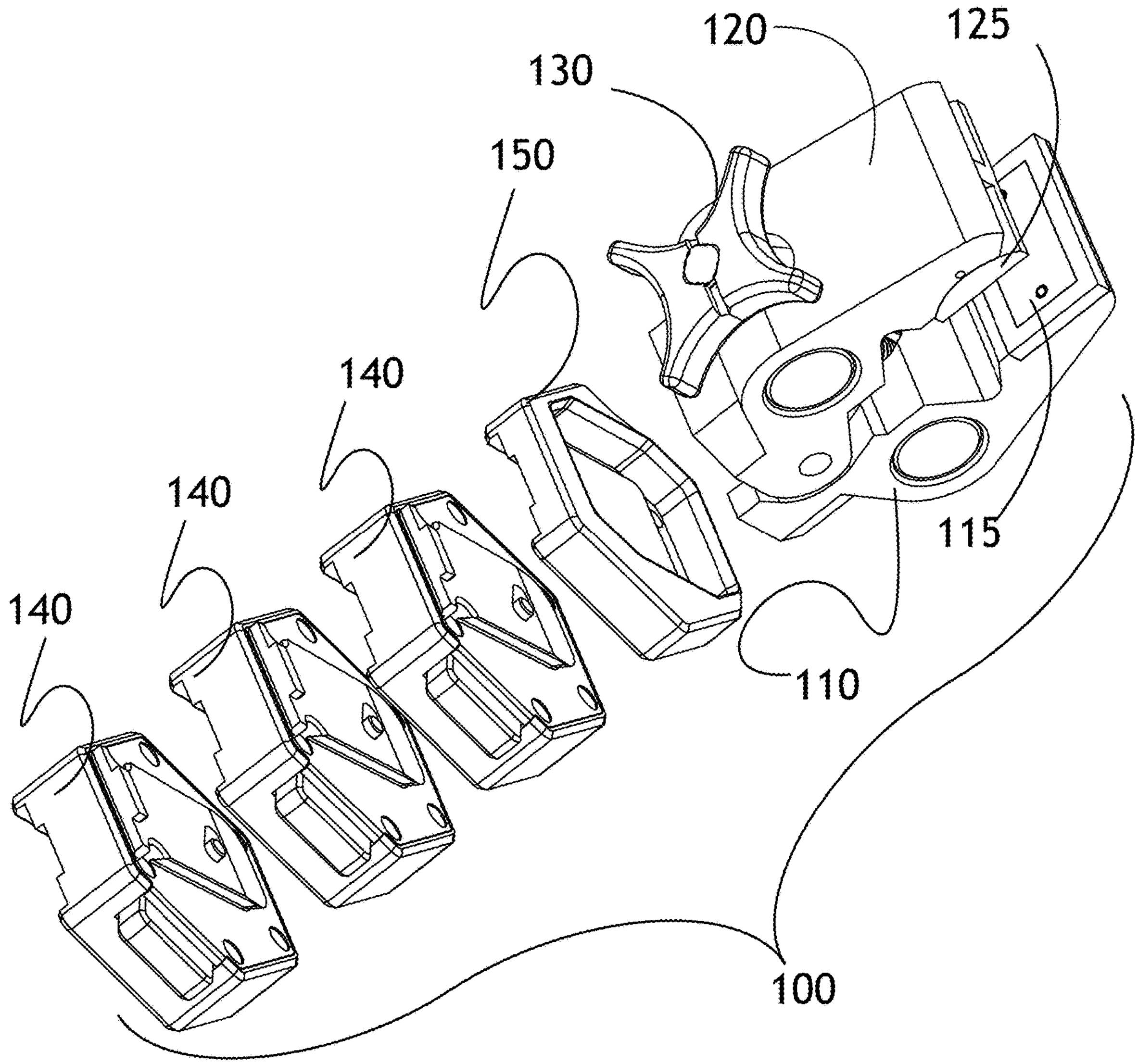


FIG. 10

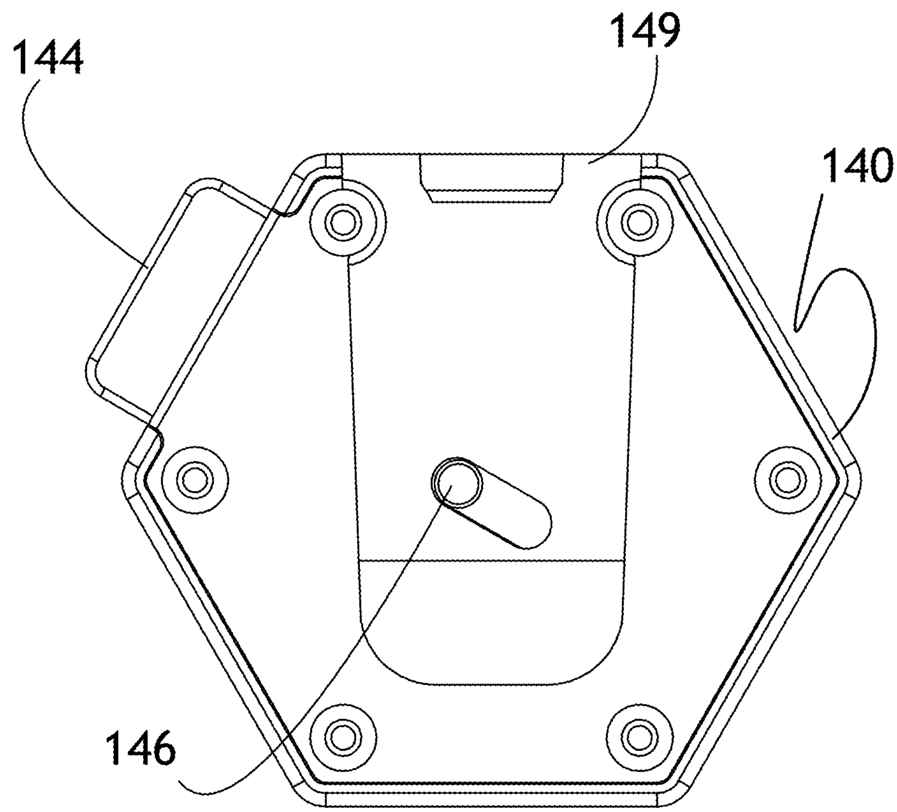


FIG. 11

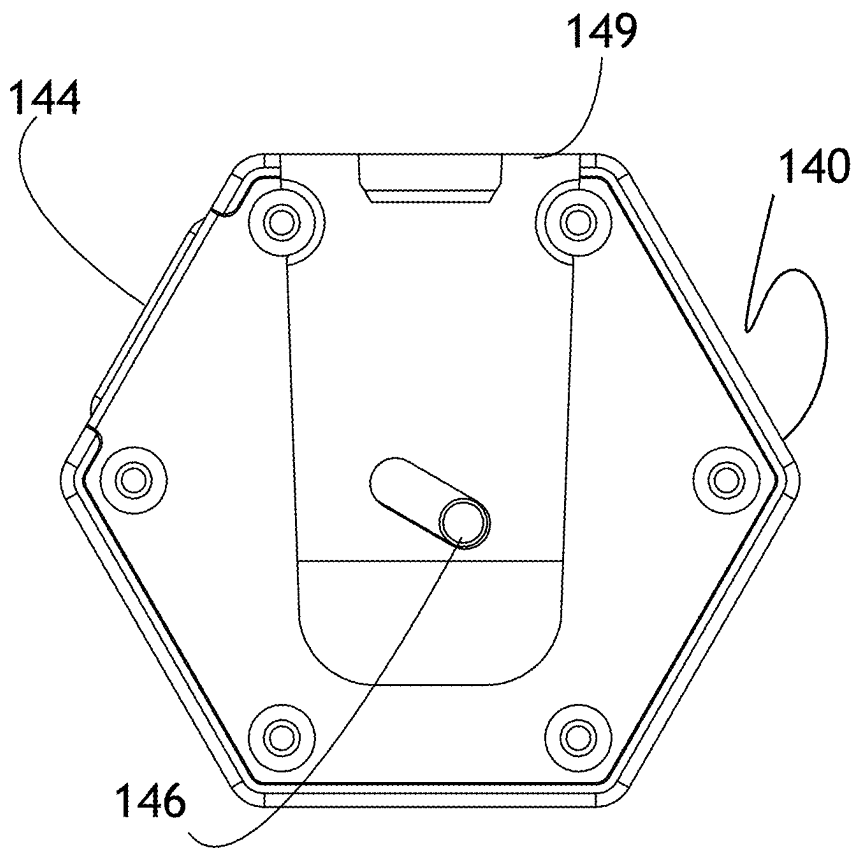


FIG. 12

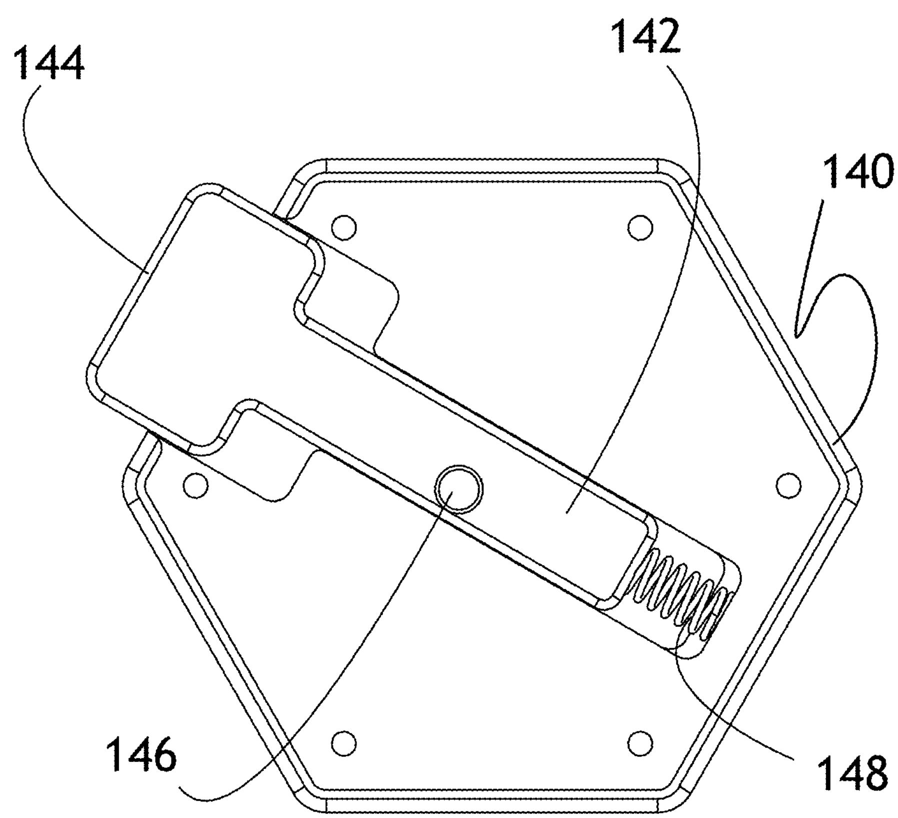


FIG. 13

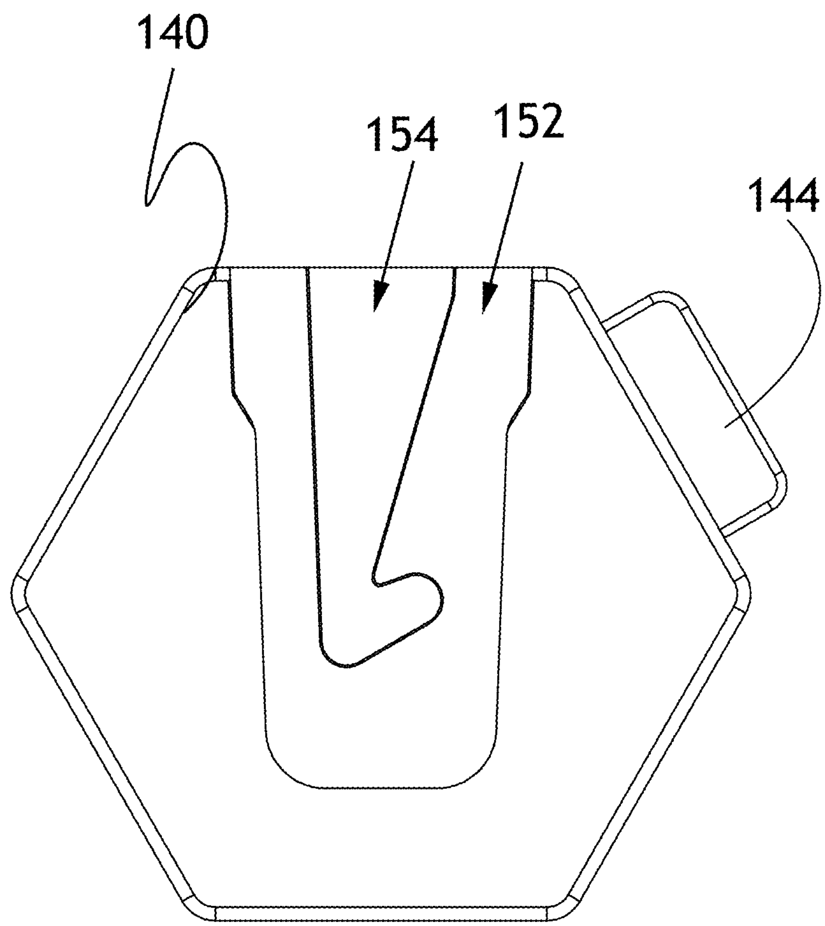


FIG. 14

ADAPTIVE RESISTANCE SYSTEM FOR PHYSICAL THERAPY AND BIOMECHANICS TRAINING

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] The present invention claims priority to prior filed U.S. Application No. 63/559,128, filed on Feb. 28, 2024 and titled TRAINING AND REHABILITATION WEIGHTS FOR RACKETS AND OTHER SPORTS EQUIPMENT, and incorporates the same by reference herein in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of physical rehabilitation and training and more particularly relates to an adaptive clamp-on weight attachable to rackets, paddles, sticks, and other equipment utilized in sporting events.

BACKGROUND OF THE INVENTION

[0003] Physical therapy is used to recover from many forms of injury, whether from accidents, on-the-job injuries, or injuries from sports. Physical therapy utilizes exercises, stretches, and other modalities to help a person recover from these injuries. Often, physical therapy may utilize sport training in a rehabilitation program. However, physical therapy tends to focus on the aftermath, rather than the conditioning necessary to avoid injury in the first place, particularly in sports or in the course of employment. This type of "pre-habilitation" can not only reduce incidence of injury but also improve performance in a sport. However, tools which can be used for both purposes are few.

[0004] Sports are a popular form of recreation, whether by playing or watching. Sports may be singles or team events, and have varied rules concerning how points are scored, how a projectile is or may be handled, even how a player may move. Many sports involve using some form of equipment, such as a paddle, racket, bat, club, or stick, to handle some form of projectile, like a ball, puck, or shuttlecock, and direct said projectile in a particular direction, as determined by the rules and strategy employed in the game. When utilizing projectile handling equipment, players develop particular moves and methods of using the equipment, again depending on the rules of the game and type of equipment. In certain sports, such as baseball or cricket, the projectile (a ball) is simply handled by hitting it as far as possible (unless strategy deems otherwise) when the ball is pitched or hurled towards the player. Golfers swing clubs at balls to hit them towards a target many yards away. In lacrosse, a team of players attempts to handle a ball by passing it between players using a specialized stick, with a specialized netting, to catch and throw the ball until a goal is scored. Likewise, in hockey, a bent hockey stick is used to pass a puck (or ball if field hockey) between players in similar fashion. Racket sports, which would include paddle sports, a ball (or shuttlecock in badminton) is volleyed between players until one misses the ball or the ball is hit out of bounds, at which time a point is scored.

[0005] In each of these, and other similar sports, handling the equipment is of paramount importance, involving many facets of a player's physiognomy to expertly operate said equipment. It is understood that the hands, wrists, fore arms, elbows, upper arms, and shoulders will all play a part in handling said equipment. The arm (including the hands) is actually a complex structure, involving thirty-two bones (when the shoulder's scapula and clavicle are included), fifty-eight muscles, and many tendons and ligaments to hold the structure together. Thirty-four muscles and twenty-seven bones are in each hand alone, while the fore arm has nineteen muscles (primarily for wrist manipulation) and two bones. As there are many aspects to the aforementioned sports which are enhanced by the proper training and strengthening of these muscle groups, many exercises and techniques have been developed for the strengthening, rehabilitation, and training of the muscles and tendons of the arm.

[0006] It should therefore come as no surprise that various forms of training techniques and equipment have been developed to assist in the training, strengthening, and rehabilitation of a player's arms, or other muscle groups. Other than what one would think of as regular "strength training," another training discipline is plyometrics. Plyometrics utilizes techniques to help muscles transition from extended to contracted form quickly, resulting in "explosive" movement. These rapid transitions can prove useful in racket and paddle sports and certain stick-utilizing sports like lacrosse, where a player's reactions need to be fast enough to keep up with or out-perform an opponent.

[0007] A common training technique is to add weight to a racket or paddle. Many devices have been made which fit around the strings making the head of a tennis racket or the basket of a lacrosse stick. These weights have the advantage of possibly using the player's own equipment, including any grip modifications, in training exercises. These structures tend to be secured in the spaces between said strings and envelop said strings to some extent. This makes these weights particularly time consuming to install or remove, inhibiting the use of the training weights for different exercises which may require different weights. It also becomes troublesome to remove the weight before using the equipment in a game. These devices are also typically device specific and cannot be used for other equipment. There then exists a need for a training weight which may be easily installed and removed, be used with various forms of sports equipment and is scalable for various training exercises.

[0008] The present invention is an adaptive resistance system utilizing a clamp and modular weights which may be affixed to many different forms of sports equipment. The clamp has an attachment system by which the modular weights may be added thereto. This system may then not only be used for conditioning and pre-habilitation, but also for physical therapy and rehabilitation. The present invention represents a departure from the prior art in that the adaptive resistance system of the present invention is designed to seamlessly integrate precision-engineered resistance-weighted modules with existing sporting equipment. At its core, it features a proven modular clamp-on weight system, engineered for easy attachment to rackets, paddles, sticks, and other athletic gear. This adaptable system not only optimizes strength conditioning but also supports rehabilitation and injury prevention, enhancing both performance and recovery in training and competitive environments. It represents a breakthrough in physical therapy innovation, offering a comprehensive solution for athletes at all levels.

SUMMARY OF THE INVENTION

[0009] In view of the foregoing disadvantages inherent in the known types of training aids, an improved easily positionable training weight may provide a training aid that meets the following objectives: that the training weight be easily manufacturable for various types of sports equipment, that it be easily positioned about an existing racket or other item of sports equipment, that use of the training weight and such positioning does not alter said racket or other sports equipment, that it presents variable weight options. As such, a new and improved training weight may comprise modular weight units attachable to a clamp adapted to fit about an item of sports equipment, such as, but not limited to, a tennis racket, pickleball paddle, ping-pong paddle, hockey stick, lacrosse stick, jai alai cesta, or other similar item of equipment, in order to accomplish these objectives.

[0010] This invention establishes a groundbreaking category of hybrid rehabilitation and prehabilitation technology by seamlessly integrating precision-engineered resistance-weighted modules with existing sporting equipment. At its core, it features a proven modular clamp-on weight system, engineered for strategic attachment to rackets, paddles, sticks, and various athletic gear. This adaptable system optimizes strength conditioning while simultaneously supporting rehabilitation and injury prevention, enhancing both performance and recovery within training and competitive environments.

[0011] The more notable features of the invention have thus been outlined in order that the more detailed description that follows may be better understood and in order that the present contribution to the art may be better appreciated. Additional features of the invention will be described hereinafter and will form the subject matter of the claims that follow.

[0012] Many objects of this invention will appear from the following description and appended claims, reference being made to the accompanying drawings forming a part of this specification wherein like reference characters designate corresponding parts in the several views.

[0013] Before explaining at least one embodiment of the invention in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangements of the components set forth in the following description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced and carried out in several ways. Also, it is to be understood that the phraseology and terminology employed herein are for description and should not be regarded as limiting.

[0014] As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the present invention. It is important, therefore, that the claims be regarded as including such equivalent constructions as far as they do not depart from the spirit and scope of the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] In order to describe the manner in which the above-recited and other advantages and features of the invention can be obtained, a more particular description of the invention briefly described above will be rendered by reference to specific example embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are therefore not to be considered as limiting of its scope, the invention will be described and explained with additional specificity and detail using the accompanying drawings.

[0016] FIG. **1** is a front elevation of a pickleball paddle with one embodiment of a training weight attached at a top position.

[0017] FIG. **2** is a front elevation of a pickleball paddle with two of the training weights of FIG. **1** attached at opposed lateral positions.

[0018] FIG. **3** is a sectional view of the pickleball paddle and training weight of FIG. **1**, taken along line III-III.

[0019] FIG. **4** is a sectional view of the pickleball paddle and training weight of FIG. **3**, with the weight clamp opened.

[0020] FIG. **5** is a perspective view of the paddle and weight of FIG. **1**, before attachment.

[0021] FIG. **6** is a side elevation of the paddle and weight of FIG. **3**.

[0022] FIG. **7** is a perspective view of the paddle and weight of FIG. **1**,

[0023] FIG. **8** is a side elevation of the paddle and weight of FIG. **1**.

[0024] FIG. **9** is an exploded view of the training weight of FIG. **1**.

[0025] FIG. **10** is an alternate view of the exploded view of FIG. **9**.

[0026] FIG. **11** is a bottom plan view of a weight, as used in the preceding Figures.

[0027] FIG. **12** is the weight of FIG. **11**, with its plunger depressed.

[0028] FIG. **13** is a bottom plan view of the weight of FIG. **11**, with its cover removed.

[0029] FIG. **14** is a top plan view of the weight of FIG. **11**.

LIST OF REFERENCE NUMERALS

- **[0030]** **100**—one embodiment of a training weight adapted for pickleball paddles;
- **[0031]** **110**—base jaw of the clamp of the training weight;
- **[0032]** **112**—accommodation groove of the base jaw;
- **[0033]** **115**—clamping surface of the base jaw;
- **[0034]** **120**—upper jaw of the clamp of the training weight;
- **[0035]** **122**—accommodation groove of the upper jaw;
- **[0036]** **123**—concave interface of upper jaw;
- **[0037]** **125**—clamping surface of the upper jaw;
- **[0038]** **127**—convex interface of upper clamping surface;
- **[0039]** **130**—attachment bolt head;
- **[0040]** **135**—attachment bolt body;
- **[0041]** **140**—modular weights;
- **[0042]** **142**—plunger
- **[0043]** **144**—plunger head/button;
- **[0044]** **146**—cam peg;
- **[0045]** **148**—spring;
- **[0046]** **149**—boss;
- **[0047]** **150**—clamp head;
- **[0048]** **152**—key way;
- **[0049]** **154**—hooked cam track;
- **[0050]** **200**—pickleball paddle;

[0051] **210**—exterior surface, or "face," of the pickleball paddle; and

[0052] **220**—pickleball paddle edge guard, or "rim."

DESCRIPTION

[0053] With reference now to the drawings, a preferred embodiment of the training weight is herein described. It should be noted that the articles "a," "an," and "the," as used in this specification, include plural referents unless the content clearly dictates otherwise. This Specification is illustrated with, and the drawings utilize, a pickleball paddle. It should be readily understood that this is merely exemplary and that, as described later, the invention may be practiced on other forms of sports equipment.

[0054] With reference to FIG. **1**, the invention may be embodied as a simple clamp structure with modular weights **100** which may be placed along the rim of a pickleball paddle **200**. FIG. **1** illustrates a placement at the top of the paddle, opposite the handle. Placement of the weight **100** is dependent upon the desires of the user, and multiple weights **100** may be used, as shown in FIG. **2** where two weights are positioned opposite each other and laterally about the paddle **200**. Weight placement allows for different exercises to be practiced with different weight resistance for different desired effects on the human anatomy.

[0055] In the depicted embodiment, shown in FIGS. **3-9**, the training aid may feature a clamp having a base jaw **110** and an upper jaw **120**. In this Specification, the terms "base" and "upper" merely differentiate between a jaw portion which may be said to be stationary by defining a frame of reference and another which may be said to be pivotably mounted thereon. No directional limitation is intended to be inferred from the terms. The upper jaw **120** is pivotably mounted to the base jaw **110** in a manner that they mutually present accommodation grooves **112**, **122** and clamping surfaces **115**, **125**, towards each other. Clamping surfaces **115**, **125** may touch when the clamp jaws are closed and may be covered with a rubberized, textured, or other treatment to create a friction-positive surface. One or both surfaces **115**, **125** may be positioned on a pivoting base to adapt to different thicknesses of clamped surfaces (inter alia, different thicknesses of rackets). Accommodation grooves **112**, **122** are present to accommodate the edge guard, or rim **220**, present in many paddles, sticks, or rackets. It should be noted that this rim **220** may or may not be present and may vary in height between rackets or sticks for the same intended use. It should also be realized that the size and shape of the clamp jaws, clamping surfaces, and accommodation grooves will vary depending upon the desired contexts of use.

[0056] As can be seen in FIGS. **3-8**, the training weight is simply attached by opening the jaws **110**, **120**, placing them about a rim **220**, and securing the jaws in a manner to prevent release. The disclosed locking means is a simple bolt **135** extending through the upper jaw **120** and anchored by some method in or on the base jaw **110**. This bolt **135** may be secured by extending through the base jaw **110** and being fastened thereto by a simple nut, or bolt **135** may be anchored within the body of base jaw **110** itself. Once the clamp is positioned over the rim **220**, the bolt **135** may be tightened by twisting its head **130** until the jaws **110**, **120** fasten about the rim **220** and the clamping surfaces **115**, **125** are securely abutting the face **210** of the paddle **200**. The bolt **135** may then be locked in any manner or may be left as is. Once secured, the user may then practice and utilize training exercises, whether generally or those developed especially for their chosen sport, as desired.

[0057] One particular feature of the training weight **100** is the availability of modular weight discs **140**, as shown in FIGS. **9** and **10**. The training weight allows use without a modular weight (as the clamp itself will add a certain amount of weight) and for the addition of one or more additional weights **140** as desired by the user. The modular weights **140** may be positioned on an attachment module **150** likewise attached to base jaw **110** and held in place by many different strategies. One strategy could be a threaded nut on an attachment bolt. Another method, as is illustrated, is the use of a cam lock system as is more clearly shown in FIGS. **11-14**. Each weight **140** may contain an internal plunger **142** with an external button **144**, biased into a default position by a spring **148**. A cam peg **146** extends through a slot in the weight's cover. The cam peg extends through a boss **149** that likewise extends to one edge of the weight, ideally not along a line defined by the plunger **142**. The boss **149** is beveled and fits within a rimmed keyway **152**, provided on the opposite side of the weight (FIG. **14**) and on attachment module **150**. Within the keyway **152** is a J-hooked cam track **154** which guides the cam peg **146**. Sliding the weight **140** along the keyway **152** biases the cam peg **146** and the associated plunger **142** against the spring **148**. Upon reaching the hook, the pressure is released and the cam peg **146** resides within the hooked end. The boss **149** and keyway **152** then secure the weight **140** from being lifted or removed in any direction but the linear direction of the boss **149**. The cam peg **146** secures the weight **146** from sliding off along the boss **149**. Removal is then accomplished by pressing the button **144** and releasing the cam peg **146**. It should be understood that the arrangement described is the best mode, but it is conceivable that the location of the boss **149** and keyway **152** could be reversed, with a boss **149** and plunger **142** located upon mount **150** instead of keyway **152**. Other means of securement may be further developed.

[0058] Additional clamping surfaces may be developed for various rackets, paddles, or sports sticks.

[0059] In addition to training for better performance, physical therapy is a potential use of the invention. The particular use of the training weights on a paddle or racket, or any particular hand-held tool (not just sports equipment), allows for targeted, low-impact exercises which can strengthen individual muscles and tendons in the arm. Likewise, using the training weight on a stick or similar device can help target muscles of the forearm and back. These exercises can then be used to help rebuild muscles in specific muscle groups and increase flexibility and help re-establish balance and coordination of a patient. It is not the purpose of this application to outline specific exercises which may be utilized for either rehabilitation or performance, but rather to describe a new tool for trainers to enhance current practices and develop other exercises which may incorporate these tools.

[0060] Although the present invention has been described with reference to preferred embodiments, numerous modifications and variations can be made and still the result will come within the scope of the invention. The embodiments described are to be considered in all respects only as illustrative and not restrictive. No limitation with respect to the specific embodiments disclosed herein is intended or should be inferred. Therefore, the scope of the invention is indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

What is claimed is:

1. A resistance system comprising:

a clamp, further comprising:

two opposed jaws;

a locking mechanism to secure the jaws in a fixed position; and

a mount for at least one modular weight;

at least one modular weight, attachable to the mount and each weight also attachable to other weights by the same means for attachment as needed to attach to the mount; and

an item of sports equipment, upon which the clamp is affixed.

2. The resistance system of claim **1**, each weight further comprising:

a boss on one surface of the weight;

an internal spring-actuated plunger with a portion extending outside a perimeter of the weight, the plunger also comprising a cam peg extending from a surface of the boss; and

a keyway on an opposite surface of the weight, made to interface with the boss of another weight and further comprising a J-hook cam track;

wherein, the boss is slid into the keyway and the cam peg interfaces with the J-hook cam track to secure weights together.

3. The resistance system of claim **2**, the mount further comprising a keyway and J-hook cam track.

4. The resistance system of claim **2**, the mount further comprising an boss and internal spring-actuated plunger with a cam peg extending from a surface of the boss.

5. The resistance system of claim **1**, the item of sports equipment being selected from the set of sports equipment consisting of: a racket, a paddle, a sport stick, and a cesta.

6. A resistance device comprising:

a clamp, further comprising:

two opposed jaws;

a locking mechanism to secure the jaws in a fixed position; and

a mount for at least one modular weight;

at least one modular weight, attachable to the mount and each weight also attachable to other weights by the same means for attachment as needed to attach to the mount.

7. The resistance device of claim **6**, each weight further comprising:

a boss on one surface of the weight;

an internal spring-actuated plunger with a portion extending outside a perimeter of the weight, the plunger also comprising a cam peg extending from a surface of the boss; and

a keyway on an opposite surface of the weight, made to interface with the boss of another weight and further comprising a J-hook cam track;

wherein, the boss is slid into the keyway and the cam peg interfaces with the J-hook cam track to secure weights together.

8. The resistance device of claim **7**, the mount further comprising a keyway and J-hook cam track.

9. The resistance device of claim **7**, the mount further comprising an boss and internal spring-actuated plunger with a cam peg extending from a surface of the boss.

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